24. VPN Basics: IPsec, WireGuard, OpenVPN

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Phase: Phase 9 — VPN, tunnels, and overlays
Estimated time: 2 days
Slug: `24-vpn-basics`

Learning objectives

- Contrast VPN types at a high level
- Read WireGuard quick configs conceptually
- Troubleshoot VPN DNS and routes
- Watch MTU/overhead issues

Prerequisites

Complete prior lessons. Prefer a Linux VM for `ip`/`ss`/`nft` examples; macOS works for many `curl`/`dig` drills.

```
# WireGuard (if installed)
sudo wg
sudo wg show
ip route | grep -E 'wg|tun|tun0'

# OpenVPN legacy style
ip addr show tun0
```

Common failures: missing routes, DNS still public, MTU blackholes, firewall blocking UDP 51820/1194.

```
ping -c 2 10.0.0.1                # internal after VPN
dig @10.0.0.2 internal.service A
curl -I https://internal.service
```

Common mistakes

- Full-tunnel VPN without need (hairpin pain)
- Split tunnel missing critical routes
- Ignoring MTU

Exercises

Work in `exercises/24-vpn-basics/`:

```
./test.sh
```

- **A:** make tests pass
- **B:** run 2–3 real commands from this lesson in your lab and note outputs
- **C:** rewrite answers from memory next day

Do not open `solutions/24-vpn-basics/` until after an honest attempt.

Self-check

1. WireGuard show command?
2. Typical WG port?
3. Why MTU drops with VPN?

Further reading

- https://www.wireguard.com/